Exploring Promising NS5 Pockets in West Nile Virus for Future Flavivirus Antiviral Drug Developments

Keywords: antiviral drug, West Nile Virus, Flavivirus,
NS5 Protein, bioinformatics, epidemiology,
protein modeling, computational biology

Dayanara Leyva
dleyv012@fiu.edu
CURIE, Department of Biological Sciences,
Florida International University

Research under the direction of:

- Jessica Liberles (Faculty lead)
- Raquel Battifora (Teaching Assistant)
- Samuel Neely (Course Coordinator)
- Nahimana Walton (PRIMER)
- Georgia De Loza (PRIMER)

Acknowledgments: The research reported in this presentation was supported by the National Institutes of Health under award number R25AI170382. The content is solely the responsibility of the authors and does not necessarily represent the views of the NIH.

**Please note this manuscript has not been finalized and is subject to change.*

Abstract:
The rapidly emerging viral threats in Flaviviridae pose increasingly growing concerns as anthropogenic activities expand the geographic range of vectors like mosquitoes and ticks. This expansion brings viruses to areas with populations unprepared for outbreaks, highlighting the need for focused efforts on treatment development to address public and global health threats. Treatments currently presented through flavivirus vaccines introduce issues in the manifestation of antibody-dependent enhancement (ADE). Currently, there are no reported treatments for West Nile Virus (WNV) in humans, and reports of ADE with pre-existing treatments, such as Japanese Encephalitis Virus vaccines. This research's main objective is to provide viable drug target candidates with strong potential for developing broadly neutralizing antiviral drugs, using WNV as a prototype virus. The focus is on antivirals to avoid potential ADE reactions and directly target the virus. Using bioinformatics tools, we highlighted the AlphaFold-generated WNV model to have two promising candidate pockets. Both are promising due to their low evolutionary rates, high model confidence, and low intrinsic disorder. A drug targeting these may have longer efficacy as the protein is not evolving quickly. Regarding future drug modeling, the selection of WNV presents as a promising prototype virus for a broadly neutralizing antiviral target against flavivirus, with it being closely related to other viruses in the family. Thus, applications can be further manipulated for other flaviviruses. Future objectives would be to pursue empirical research while navigating possible ADE reactivity.

Introduction

The National Institute of Health reports that up to 400 million people are infected by flaviviruses annually and globally (Pierson & Diamond, 2020). The virus taxon of Flaviviridae encompasses viruses that are recognized for their positive-strand RNA. Within the taxon, Flavivirus is one of the more recognizable virus genera that encompasses Dengue 1 – 4, Zika, West Nile virus (WNV), and many more. It is essential to note that the genus itself, Flavivirus, viral pathogenicity poses a further threat to global health considering their extensive infection rates and vectors of arthropods, such as ticks and mosquitoes, which make the spread of these viruses near impossible to eradicate. Furthermore, it poses increasing public health concerns as these vectors expand their geographic range due to anthropogenic activities. This also emphasizes that infection of this virus is not localized but a widespread and global issue and the need for research to prevent outbreaks.

West Nile Virus was traced to origins in Uganda, and concerns grew as the first cases began to be documented in the United States, previously unexposed to this virus. This amplified concerns regarding vector-transmitted viruses and ecological niches changing. West Nile Virus is a particularly interesting flavivirus and the subject of this study due to the high reports of it causing neuroinvasive diseases. The CDC reported in data collected in 2021 (Fagre, A.C., 2023) that of the arboviral disease cases reported, 90% were neuroinvasive, and WNV was found to be responsible for 96% of the neuroinvasive cases. It is also interesting to note there is a WNV vaccine for equines, but not yet for humans.

It is also important to note key characteristics of Flavivirus, such as the varying evolutionary rates among other species and the disease relationship amongst one another. For this reason, a highly conserved protein, when setting an evolutionary background, would be the best target to neutralize other species. Along with targeting an important function to completely disrupt the virus infection cycle. Considering this, the research performed focuses on the protein NS5 due to the fact that the protein plays a vital role in protein replication and RNA capping

**Please note this manuscript has not been finalized and is subject to change.*

(Fajardo et al., 2020). Essentially, NS5 is the ideal target in this research as it has high modeling certainty, a larger amount of possible drug pocket targets, and relatively low percentage levels of disorder. The protein is also an allosteric site that is responsible for conformational change and the possibility of further infection (Keating et al., 2013).

When further exploring research of these viruses, specifically into the high infection rates, empirical research for possible drug treatment is already underway. However, experiments have yielded results that indicate antibody-dependent enhancement (ADE) with vaccines. The antibody levels when someone has been infected by one flavivirus will increase antibodies that may facilitate easier infection of other flaviviruses that will manipulate the antibodies to more easily infect and replicate within a host (Saito et al., 2016). For this reason, this research focuses on antiviral drug targeting and not on developing vaccines that can cause these ADE reactions.

The objective of this study is to find an optimal protein pocket for antiviral drugs to target within the Flavivirus genus by considering the proteins' most influential traits: protein evolutionary rate and conservation along evolving structure, protein structure modeling confidence, protein disorder percentage, and the evidence of feasible druggable pockets. Data comparison in the evolutionary context along WNV will aid in furthering the discovery of universally effective anti-viral drugs that can be utilized across different Flavivirus strands even as they evolve.

Methods

2.1 Protein Family Reconstruction – Sequence Dataset, MSA & Tree. The West Nile Flavivirus protein NS5 query sequence was used in the BLAST program database (accession number: YP_001527887.1). The eighteen best hits were selected in the research to create the data set. This data set was then used to create the multiple sequence alignment (MSA) file; generated using the MUSCLE default algorithm in Jalview (Waterhouse, 2009) with default settings through webservices (Troshin, 2011). IQ-Tree generated visualization of this data through a phylogenetic tree with best-fit model LG+I+G4 and sh-LRT branch testing. The phylogenetic tree's final visualization being mid-pointed, rooted in Figtree, and color-coded by vectors (Figure 1).

2.2 Evolutionary Rate Analysis – Running Rate4Site and Rate4Site Parsing. The alignment sequences (MSA) were generated and processed utilizing Rate4Site to obtain the protein's evolutionary context. The output was used in Jupyter from Python, to run a script which yielded line plots, heat maps, and a boxplot. Using a parsing script, relevant information was extracted for further analysis.

2.3 Predictive Structure Modeling. Colabfold utilizing AlphaFold2 (Jumper, 2021) generated the WNV NS5 protein structure modeling information; creating the protein database (PDB) file information (Mirdita, 2022). ColabFold, which predicts monomeric protein structures using AlphaFold 2, typically generating an output with five models per protein ranked by pLDDT (confidence) score. The following parameters were used to determine the best AlphaFold model: model confidence (pLDDT), secondary structure, and evolutionary rate. Separate sessions were concluded for each of the parameters in a stepwise fashion. The first generated a 3D model with PyMOL (Schrödinger, 2020). The second PyMOL session included one colored by evolutionary rates (Figure 4A) and another colored by b-factors/confidence (Fig 4B). The rank 1 PDB file of the 3D protein was used for each of these renderings.

2.4 Structural Analysis – Extract structural features from 3D model, map rates to 3D, disorder prediction. Extracted model confidence and secondary structure per residue of the previous output was compiled into an evolving protein datasheet. Following this- the NS5 WNV Fasta file of the query sequence was run through IUPRED, with default settings, to generate disorder prediction. This was completed by inputting the NS5 WNV FASTA file of the query sequence and run with default settings.

2.5 Predict Pockets based on Prototype Protein – Pocket Inspection. PockDrug predictions were then created. Utilizing rank 1 PDB file of the protein modeling, outputs were organized by highest to lowest druggability. This yielded a list of the best pockets, which were selected following the parameters that there were at least 10 residues and a druggability score of >0.5.

2.6 Make SQL Database. The Structured Query Language program was then used to format all the needed information collected about NS5 of WNV. The previous information was compiled into results that were extracted from NS5 and kept on an Excel for site data, into DB Browser. The previous Excel output containing residue site database information was also imported into DB Browser. From this, the rates, disorder, and confidence of each site were extracted specifically from the most promising pockets.

2.7 Pocket Analysis. The residue site database was manually marked with pockets of interest; the residues that correlated with pockets of interest were then imported into Spyder. Through Spyder, the pocket rates were analyzed with the Shapiro-Wilks test – which yielded pocket normalcy results. Following the result of this test, a Mann-Whitney U test was conducted and resulted in a comparison of pockets with resulting p-values. This statistical data analysis process was repeated an additional two times for residues disorder, and confidence. This was used to develop a chart for NS5 residue specific protein information.

Results

3.1 Evolutionary Analysis

3.1.1 Protein Family Analysis

The evolutionary context was developed with WNV. The virus' corresponding query generated the best results for similar viruses. The program BLAST database gave results for the query that had subsequent minimum and maximums of the following: query cover 98%-100%, e-value 0.0-0.0, percent identity 44.97%-100%, and accession length of 643-3451. Of these results, the first 18 when sorted by e-value, percent identity, and query cover, were selected. This data set includes results from mosquito-borne flavivirus (MBFV) with *Aedes spp*, *Aedes* outgroup, and *Culex spp* subgroups. There were also results included from Tick-Borne flavivirus (TBFV) and one within MBFV but unidentified subgroup. The information was taken to generate visualizations of evolutionary rates. These relationships were then visualized via a phylogenetic tree. The tree was midpoint rooted and color-coded concerning the known vectors (Figure 1).

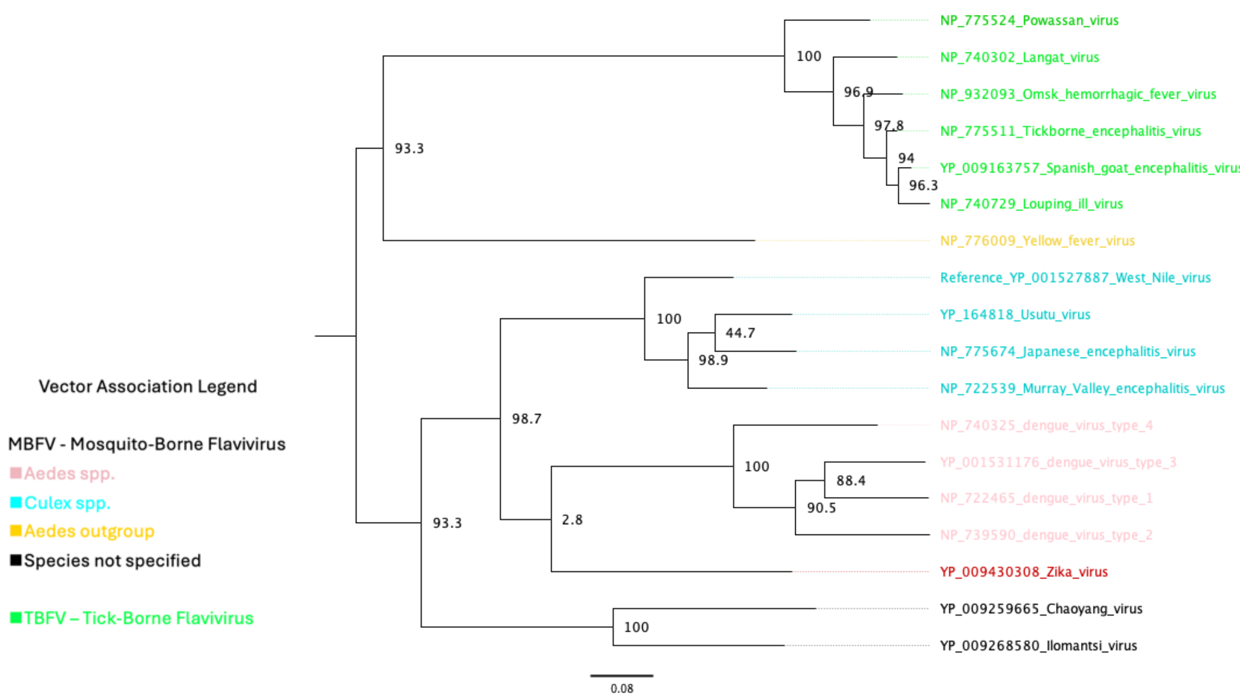


Figure 1. Phylogenetic Tree. Tree visualized with reference to WNV with the top 18 best hits of the BLAST results. The results are color-coded by the reference as above: Tick-Borne Flavivirus in green, Aedes outgroup in yellow, Culex spp. in cyan blue, Aedes spp. in pink, and unspecified species vectors in black.

3.1.2 Evolutionary Rates

The Jupyter output was used alongside a script that analyzed the information for evolutionary rates. The resulting Rate4Site generated graph that creates visualizations of evolution in areas with high peaks while also providing low peaks that represent low evolutionary rates.

**Please note this manuscript has not been finalized and is subject to change.*

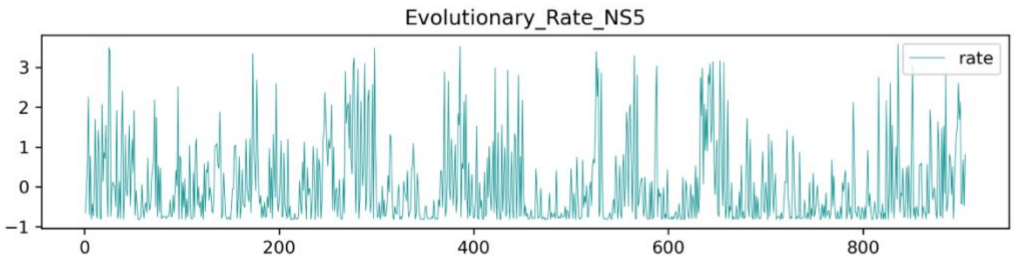
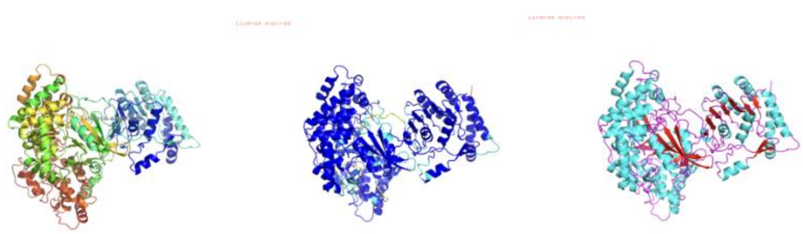


Figure 2. NS5 Evolutionary rates graph. It is important to note the length and large size of NS5, which is seen below as the residues reach past 800. Varying levels of evolution are to be expected in such a large protein. There is a variation of low and high peaks, but notable low peaks are from residues ~300 to ~370, ~450 to ~550, and ~650 to ~800.

3.2 Structural Analysis

3.2.1 Protein Structure modeling

The figure below provides a 3D model of protein NS5 in the West Nile Virus visualized through three viewings. The first being secondary structure which are each colored from N-terminal to C-terminal following the order of sequence for residues from violet, blue, green, yellow, orange, to red. Bearing this in mind, the chain begins on the right (relative to the figure orientation below). The second figure (B) is the protein in the same orientation and now colored by model confidence- dark blue representing very high confidence. It is worthy to note the overall high confidence modeling the program resulted in. The final viewing of the protein (C) was visualized via secondary structures- cyan representing helixes, sheets colored in red, and loops in magenta.



(A) Chain **(B) Confidence** **(C) Secondary Structure**

Figure 3. **(A)** NS5 West Nile Virus rank 1 visualized by secondary structure. **(B)** NS5 West Nile Virus rank 1 visualized by model confidence; dark blue: very high confidence (pLDDT > 90), light blue: high confidence score (90 > pLDDT > 70), yellow: low (70 > pLDDT > 50), orange: very low (pLDDT < 50) (Piovesan, 2022)). **(C)** NS5 West Nile Virus rank 1 visualized by secondary structure with the following color coding: cyan represents helixes, red represents sheets, and magenta represents loops.

3.2.2. Structural Analysis

A pattern can be observed in the NS5 protein of WNV of evolutionary rate. The model shows high confidence in the helix components of the protein and the more exposed parts of the helixes appear to have slightly higher evolutionary rates than the ones more towards the center of the protein.

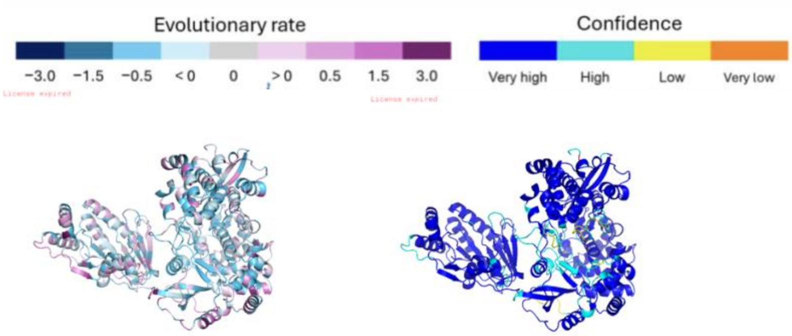


Figure 4. Protein structure of NS5 AlphaFold models in PyMOL. **A)** AlphaFold model visualized via evolutionary rate (low being dark blue with a gradient to dark purple that represents high evolutionary rate). **B)** AlphaFold model colored via confidence according to pLDDT (dark blue representing areas of high confidence, cyan with slightly lower confidence, yellow for low confidence, and orange for very low confidence).

3.3 Pocket Analysis

3.3.1. Pocket Prediction

The result of extracting site-specific information yielded various promising drug pockets, as follows: 29, 26, 32, 18, 8, 10, 45, 7, 2, 13, 37, 4. These results were sorted based on druggability and thus pocket 29 has the highest reported druggability and 4 the lowest (relative to one another as the parameters set yielded results with at least 10 residues and a druggability score of >0.5). For clarity, the pockets were renamed based on the characteristics mentioned above seen in Figure 5. These pockets were then each compared against one another based on evolutionary rates, confidence, and disorder percentages through Mann-Whitney U tests that showed significant differences in certain pocket comparisons.

Pockets Compared	P-value	Significant	Test Used
0 v 1	0.046	YES	MANN-WHITNEY U
0 V 2	0.130	NO	MANN-WHITNEY U
0 V 3	0.023	YES	MANN-WHITNEY U
0 V 4	0.979	NO	MANN-WHITNEY U
1 V 2	0.926	NO	MANN-WHITNEY U
1V3	0.683	NO	MANN-WHITNEY U
1V4	0.056	NO	MANN-WHITNEY U
2V3	0.621	NO	MANN-WHITNEY U
2V4	0.255	NO	MANN-WHITNEY U
3V4	0.107	NO	MANN-WHITNEY U

Figure 6A. Statistical NS5 pocket comparison of evolutionary rates through Mann-Whitney U test.

Pockets Compared	P-value	Significant	Test Used
0 v 1	0.099	NO	MANN-WHITNEY U
0 v 2	0.751	NO	MANN-WHITNEY U
0 v 3	0.406	NO	MANN-WHITNEY U
0 v 4	0.146	NO	MANN-WHITNEY U
1 v 2	0.172	NO	MANN-WHITNEY U
1 v 3	0.289	NO	MANN-WHITNEY U
1 v 4	0.890	NO	MANN-WHITNEY U
2 v 3	0.553	NO	MANN-WHITNEY U
2 v 4	0.222	NO	MANN-WHITNEY U
3 v 4	0.393	NO	MANN-WHITNEY U

Figure 6B. Statistical NS5 pocket comparison of confidence through Mann-Whitney U test.

Pockets Compared	P-value	Significant	Test Used
0 v 1	0.016	YES	MANN-WHITNEY U
0 v 2	0.805	NO	MANN-WHITNEY U
0 v 3	0.805	NO	MANN-WHITNEY U
0 v 4	0.451	NO	MANN-WHITNEY U
1 v 2	0.005	YES	MANN-WHITNEY U
1 v 3	0.006	YES	MANN-WHITNEY U
1 v 4	0.043	YES	MANN-WHITNEY U
2 v 3	1.000	NO	MANN-WHITNEY U
2 v 4	0.781	NO	MANN-WHITNEY U
3 v 4	0.751	NO	MANN-WHITNEY U

Figure 6A. Statistical NS5 pocket comparison of disorder percentages through Mann-Whitney U test.

3.3.2. Pocket Analysis

Decidedly, pockets 0 through 4 were chosen to further pursue comparison considering the aforementioned parameters. These were used to produce statistical visual representations of the data sets. These visualizations show pockets 0 and 4 as having high confidence, low evolutionary rate, and extremely low disorder percentages.

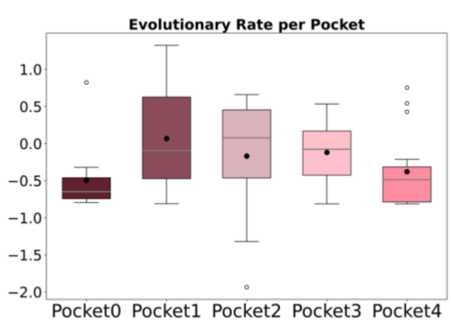


Figure 7A. Boxplot showing rates per pocket.

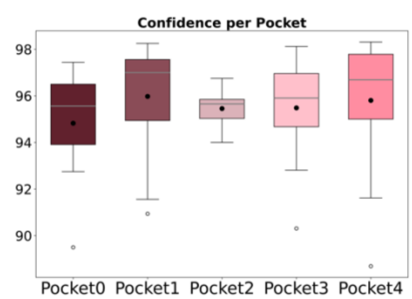


Figure 7B. Boxplot showing confidence (pLDDT) per pocket.

**Please note this manuscript has not been finalized and is subject to change.*

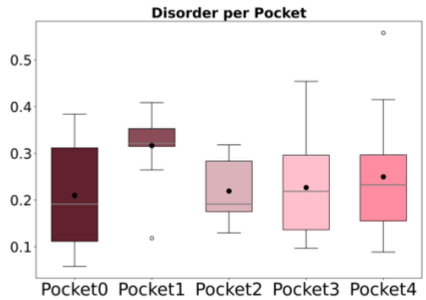


Figure 7C. Boxplot showing disorder per pocket.

3.3.2 Pocket Visualizations

The previous visualizations were used in combination to information from the created SQL database with residue numbers of the corresponding pockets of interest in order to create 3D structures with each pocket visible.

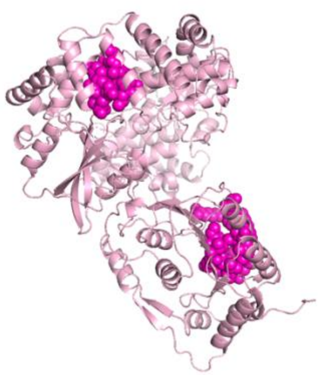


Figure 8. Protein NS5 in light pink and pocket 0 (org. 29) and pocket 4 (org. 8) visualized through 3D modeling in PyMOL.

Discussion

The results of this study reaffirm previous mentions of the NS5 protein as a promising drug target in the context of viral drug design when discussing Flavivirus. Using West Nile Virus as the prototype virus in this study, the BLAST results yielded the 17 best hits that revealed high similarity. These results indicate evolutionary relationships that provide context to further understand the conservation of the NS5 function in the Flavivirus genus. This was via the generated tree, which displayed high percentages of confidence in the relationship between viruses (displayed on the node length). It also showed low degrees of divergence, as indicated by the low genetic changes of 0.08. Furthermore, visualizations extracted from the MSA showed low evolutionary rates in the protein. AlphaFold modeling provides ranks of the protein modeling predictions- ultimately, the research was conducted with only the rank 1 model for further research as it represents the most accurate 3D structure. This evolutionary context laid the foundation for 3D visualization of the protein, colored according to evolutionary rate and confidence, which displayed favorable sites. It should be noted that comparisons of these visualizations revealed high overall conservation. Dark blue sections that represent high conservation were reflected in cyan helices and red sheets. While loops reveal high conservation in light blue, indicating slightly less conservation than the dark blue helices. The best pockets were then selected based on the parameters of containing at least 10 residues and a draggability of over 0.5, and they were used to assess the best drug targets. Having many residues helps to create drug targets in that it provides more possibilities for it to bind to. The druggability scores of over 0.05 also improve the creation of antiviral drug treatment in that statistical analysis predicts a higher binding affinity for the said drug to specific pockets.

The NS5 protein is a promising target used in this research due to its essential role in viral replication, capping, and conservation across the Flavivirus genus. The aim of disrupting this protein can lead to cascading effects in protein function- one of which includes the nucleoside and allosteric inhibitors of NS5-encoded RNA-dependent RNA polymerase (RdRp) activity (Pierson & Diamond, 2020). This interrupts the NS5 RNA replication cycle. NS5 is also responsible for RNA capping that protects the genome ends. By interfering with these functions through a drug, there is the possibility of reducing infections and subsequent severity during the replication phase of the virus. Interfering would mean undermining the structural integrity of the protein infection and disrupting RNA-dependent RNA polymerase processes that are essential to viral replication. Disrupting the capping process introduces instability in the protein that may slow or stop virus protein growth- inhibiting further viral replication.

The insights gained from the study on protein pocket NS5 in the West Nile virus have significant implications for drug development and design, highlighting promising pockets. The initial reasoning for picking NS5 as the drug target was due to it having high conservation of the overall protein which made it an ideal target, as reduced variability in related viruses suggests a lower likelihood of rapid adaptation to the drug. This enhances the drug's reliability for long-term outcomes. Following this, the generated modeling through PDB structures indicated high confidence in the proteins' 3D structure (Figure 3B)- meaning it would yield close to exact real-world models and creating drugs with this in practice increases the likelihood of the treatment proving effective. The other important facet of this research was dedicated to pocket comparisons with specific residues in mind. All pockets listed in Figure 5 proved promising characteristics, however, the top five of said pockets were ultimately chosen to continue further examination. The results yielded promising pockets the most notable being pockets 0 (originally pocket 29) and pocket 4 (originally pocket 8). Pocket 0 residues recorded in positions 490, 493, 548, 551, 552, 554, 570, 574, 575, 607, 611. Pocket 4 residues recorded in positions 104, 132, 146, 147, 148, 156, 159, 160, 162, 163, 164, 180, 181, 182, 183.

In these graphically presented data sets, there were two that stood out the most: pockets 0 and 4. These pockets had a low evolutionary rate- meaning antiviral drug treatments would hold longer efficacy. Pockets 2 and 3 were excluded for further evaluation as comparison with the other promising targets showed they had a higher evolutionary rate. Following this, the confidence of each pocket was compared against one another. Pockets 0 and 4 had high modeling confidence, which again, means antiviral drug creation can be conducted with more assurance that it will be a viable target in the physical structure. Finally, the pockets also had very low disorder percentages. Pocket 1 was not considered as the best candidate due to its higher relative disorder percentage seen in Figure 7C. This is especially important to contribute to drug creation in that there is reduced concern of protein modeling notwithstanding physical experimentation.

Another extremely important visualization was that of the pockets in the protein. Both these pockets (0 and 4) are easily accessible to a drug as they are on the protruding ends of the NS5 protein and are large in residues (Figure 8), which aids in providing a drug with more possible sites to bind to. Understanding these as the more favorable pockets, it is especially interesting to emphasize their accessible positions as NS5 is explained to be an allosteric inhibitor site. Where the disruption of these pockets could lead to reduced activity and, subsequently, less virality of the West Nile Virus.

While this research provides valuable insights into further drug development targeting the NS5 protein for Flavivirus, several limitations must be acknowledged, including the cross-reactivity of other viruses when considering administered drugs. One such important factor is understanding the levels of antibodies; insufficient antibody levels may increase susceptibility to other Flavivirus infections, as cross-reactivity may lead to an inadequate immune system response (Salem et al., 2024). Another important limitation is recognizing that this approach is entirely theoretical and based on computer programming predictions. A promising direction for future studies would be to investigate the remaining six pockets identified by PockDrug, as they could offer valuable insights. Empirical research deriving from modifications of a pre-existing medication pertaining to the Flaviviridae taxon, such as Japanese Encephalitis vaccines, could prove a positive background subject in advancing the antiviral drug treatments that exist. This vaccine that targets Japanese encephalitis could be beneficial to modify towards West Nile Virus as they are closely related and thus increase the possibility of efficacy while reducing severe ADE. As mentioned, it is also important to conduct these empirical research trials to further understand what ADE may come as a result of drugs that target NS5 of West Nile Virus.

References:

Fajardo, T., Sanford, T. J., Mears, H. V., Jasper, A., Storrie, S., Mansur, D. S., & Sweeney, T. R. (2020). The flavivirus polymerase NS5 regulates translation of viral genomic RNA. Nucleic Acids Research, 48(9), 5081–5093. https://doi.org/10.1093/nar/gkaa242

Fagre, A. C., Lyons, S., Staples, J. E., & Lindsey, N. (2023). West Nile virus and other nationally notifiable arboviral diseases — United States, 2021. MMWR Morbidity and Mortality Weekly Report, 72(34), 901- 906. https://doi.org/10.15585/mmwr.mm7234a1

Keating, J. A., Bhattacharya, D., Lim, P.-Y., Falk, S., Weisblum, B., Bernard, K. A., Sharma, M., Kuhn, R. J., & Striker, R. (2013). West Nile virus methyltransferase domain interacts with protein kinase G. Virology Journal, 10, 242. https://doi.org/10.1186/1743-422X-10-242

Jumper, J. et al. "Highly accurate protein structure prediction with AlphaFold." Nature, 596, pages 583–589 (2021). DOI: 10.1038/s41586-021-03819-2

Keating, J. A., Bhattacharya, D., Lim, P.-Y., Falk, S., Weisblum, B., Bernard, K. A., Sharma, M., Kuhn, R. J., & Striker, R. (2013). West Nile virus methyltransferase domain interacts with protein kinase G. Virology Journal, 10, 242. https://doi.org/10.1186/1743-422X-10-242

Mirdita, M. et al. "ColabFold: Making protein folding accessible to all." Nature Methods, 19, pages 679–682 (2022). DOI: 10.1038/s41592-022-01488-1

Pierson, T. C., & Diamond, M. S. (2020). The continued threat of emerging flaviviruses. *Nature Microbiology*, *5*(6), 796–812. https://doi.org/10.1038/s41564-020-0714-0

Piovesan D, Monzon AM, Tosatto SCE. Intrinsic protein disorder and conditional folding in AlphaFoldDB. Protein Sci. 2022 Nov;31(11):e4466. doi: 10.1002/pro.4466. PMID: 36210722; PMCID: PMC9601767.

Saito, Y., Moi, M. L., Takeshita, N., Lim, C.-K., Shiba, H., Hosono, K., Saijo, M., Kurane, I., & Takasaki, T. (2016). Japanese encephalitis vaccine-facilitated dengue virus infection-enhancement antibody in adults. *BMC Infectious Diseases*, *16*(1), 578. https://doi.org/10.1186/s12879-016-1873-8

Schrödinger, L., & DeLano, W. (2020). PyMOL. Retrieved from http://www.pymol.org/pymol

Troshin PV, Procter JB, Barton GJ (2011) Java bioinformatics analysis web services for multiple sequence alignment–JABAWS:MSA. *Bioinformatics* **27** 2001-2002. (doi:10.1093/bioinformatics/btr304)

Waterhouse AM, Procter JB, Martin DMA, Clamp M, Barton GJ (2009) Jalview Version 2 - A multiple sequence alignment editor and analysis workbench. *Bioinformatics* 25 1189-1191. (doi:10.1093/bioinformatics/btp033)

Supplemental Material

NS5 Drug Pockets Original vs Renamed Chart

The pockets were all initially sorted based on the highest reported druggability, and narrowed down through the selection of having at least 10 residues and a druggability score of >0.5.

Pocket Original Name	New Name
pocket29_atm.pdb	pocket0_atm.pdb
pocket26_atm.pdb	pocket1_atm.pdb
pocket32_atm.pdb	pocket2_atm.pdb
pocket18_atm.pdb	pocket3_atm.pdb
pocket8_atm.pdb	pocket4_atm.pdb
pocket10_atm.pdb	pocket5_atm.pdb
pocket45_atm.pdb	pocket6_atm.pdb
pocket7_atm.pdb	pocket7_atm.pdb
pocket2_atm.pdb	pocket8_atm.pdb
pocket13_atm.pdb	pocket9_atm.pdb
pocket37_atm.pdb	pocket10_atm.pdb
pocket4_atm.pdb	pocket11_atm.pdb

Figure 9. *NS5 WNV protein pockets were renamed 0 through 11.*

Proposed drug targets: Pockets 0 and 4 3D Protein Visualization

One point briefly discussed was the accessibility to the pockets a drug would have when examining pockets 0 and 4 as targets. Below are each of the pockets visualized through colorings: rates, confidence, and disorder percentages.

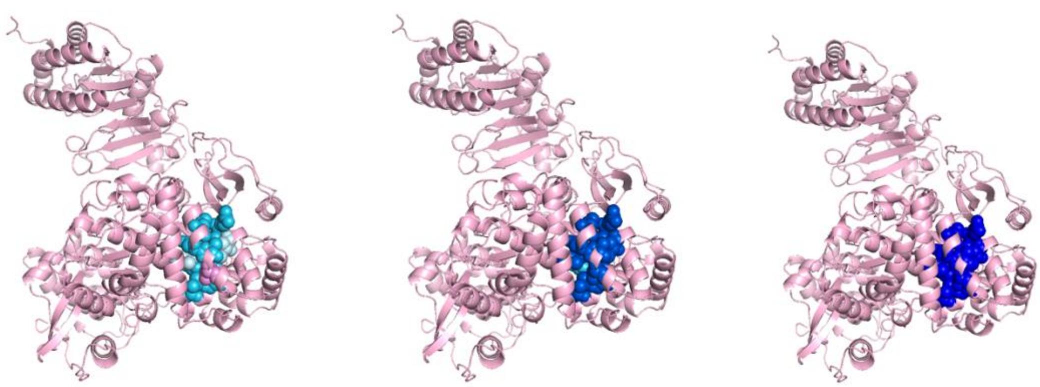


Figure 10. *3D Visualizations of Pocket 0 with the pocket's evolutionary rate, confidence, and disorder. The evolutionary rate of the pocket being depicted on a gradient scale with dark blue representing low evolutionary rate and dark purple representing high evolutionary rates. Confidence is shown through the following scale of dark blue: very high confidence (pLDDT > 90), light blue: high confidence score (90 > pLDDT > 70), yellow: low (70 > pLDDT > 50), orange: very low (pLDDT < 50) (*Piovesan, 2022). The disorder percentage is colored by blue, which indicates very low disorder, and red which indicates high disorder.

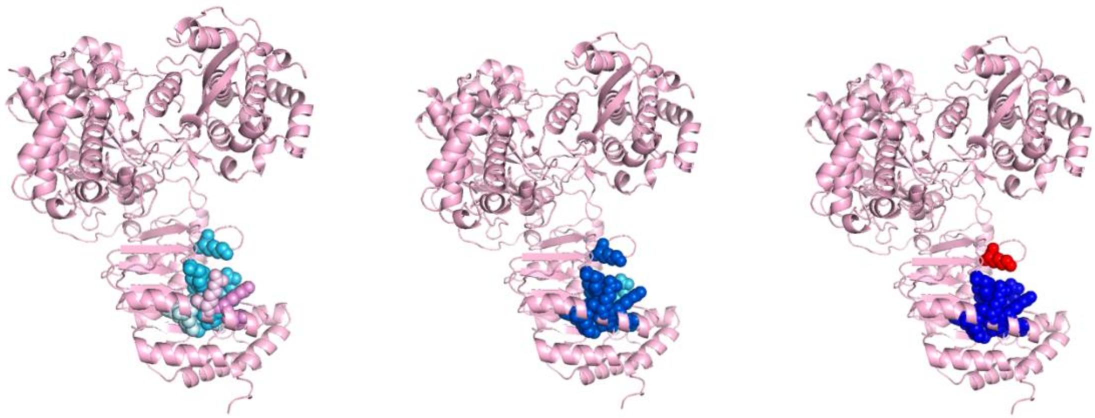
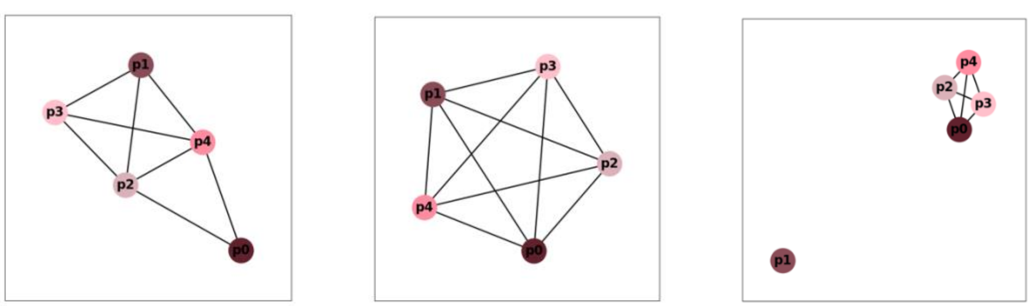


Figure 11. *3D Visualizations of Pocket 4 with the pocket's evolutionary rate, confidence, and disorder. The evolutionary rate of the pocket is depicted on a gradient scale with dark blue representing a low evolutionary rate and dark purple representing high evolutionary rates. Confidence is shown through the following scale of dark blue: very high confidence (pLDDT > 90), light blue: high confidence score (90 > pLDDT > 70), yellow: low (70 > pLDDT > 50), orange: very low (pLDDT < 50) (*Piovesan, 2022). The disorder percentage is colored by blue, which indicates very low disorder, and red, which indicates high disorder.

Pockets Network Comparison

The best pocket when comparing via the composite figure and examining their relationship is based on the evolutionary rate is 0 at around –0.5. The very close second ranking would fall to pocket 4 which is around –0.4. It is important to note that all pockets hold low evolutionary rates; all falling under 0.5 (significant difference between 0 v 1, and 0 v 3 that have medians at 0). The next characteristic evaluated was confidence, in which case all pockets exhibited high confidence. Finally, the pockets were compared by the disorder- one notable difference in the disorder of pocket 1 is significantly different than the others. In this case, all other pockets had low disorder and pocket 1 had high disorder in comparison.



(A) **(B)** **(C)**

Figure 12. *Visual comparisons of NS5 pockets by networks.* ***A)*** *Top five pockets compared by rates.* ***B)*** *Top five pockets compared by confidence.* ***C)*** *Top five pockets compared by the disorder.*

**Please note this manuscript has not been finalized and is subject to change.*